

From Particles to Waves: Coupling Crack Propagation in Snow in Spectral Elements and Dynamic Snow Modelling to Detect Cracks in DAS Data

1 Nomenclature

1.1 List of Abbreviations

DAS : Distributed Acoustic Sensing

MPM : Material Point Method

DEM : Discrete Element Method

FE, FEM : Finite Element Method

PST : Propagation Saw Test

1.2 List of Symbols

Symbols, definition and unit

t_o : Subfault onset time [s], e.g. $t_o = (x - x_{\text{start}})/v_{\text{rupture}}$.

x, x_{start} : Horizontal coordinate and rupture starting position [m].

v_{rupture} : Prescribed rupture propagation speed [m s^{-1}].

t_{rise} : Rise time of a subfault/source [s].

Δx : Subfault width or discretisation length in x [m].

x_{slip} : Prescribed slip on a subfault [m].

M_0 : Scalar seismic moment per subfault [N m].

m_{ij} : Moment tensor components [N m].

r_{mode} : Mode mix parameter [dimensionless].

l_{cz} : Cohesive process (fracture) zone length [m].

τ_s : Shear strength of interface [Pa].

l_{ch} : Irwin characteristic length [m].

λ_{min} : Shortest wavelength to resolve [m].

$v_{\text{min}}, v_{\text{max}}$: Minimum and maximum relevant wave speeds in the model [m s^{-1}].

Δt : Time step [s] for explicit time integration.

ξ : Normalized opening [dimensionless].

δ : Local opening (separation) displacement across the interface [m].

δ_c : Mixed-mode critical opening [m] used to nondimensionalize openings.

$\delta_{n,c}, \delta_{s,c}$: Single-mode critical openings for Mode I (normal) and Mode II (shear) [m].

$T(\xi)$: Cohesive traction as a function of normalised opening [Pa or N m⁻²].

T_{peak} : Peak cohesive traction [Pa].

$\phi(\Delta)$: Mixed-mode cohesive potential [J m⁻²] with $\Delta = (\Delta_n, \Delta_t)$.

Δ_n, Δ_t : Normal and tangential separations [m].

δ_n, δ_t : Normal and tangential length scales in the cohesive potential [m].

q : Coupling parameter in the mixed-mode cohesive potential [dimensionless].

ϕ_n, ϕ_t : Work (energies) of separation for pure normal and tangential modes [J m⁻²].

$\sigma_{\text{max}}, \tau_{\text{max}}$: Peak normal strength and peak shear strength [Pa].

σ_n, τ_s : Normal and shear strength [Pa].

G_I, G_{II} : Fracture energies for Mode I and Mode II [J m⁻²].

v_s, v_p : Pressure and Shear seismic wave speeds [m s⁻¹].

ρ : Mass density [kg m⁻³].

μ, λ : Lamé parameters [Pa].

E : Young's modulus [Pa].

$l_{\text{crack}}, \mathbf{x}_0$: Local crack process length and initial crack front position [m] used in local coordinate definitions.

∇ : Component-wise partial derivative operator (nabla).

θ : Reference coordinate system in respect to mesh elements used in SALVUS.

w :: Test function used to obtain the weak form of FE equations [dimensionless].

G : Model domain [dimensionless].

V : Galerkin space [dimensionless].

u : Scalar displacement potential [dimensionless].

ϕ : Set of functions which approximate solution of w or u [dimensionless].

F : Forcing term of elastic wave equation.

M : Mass term of elastic wave equation.

S : Stiffness term of elastic wave equation.

B : Boundary term of elastic wave equation.

∂ : Partial derivative in respect to one variable.

2 Introduction for Thesis

Snow avalanches are a major natural hazard in mountainous regions; they endanger human life as well as infrastructure (Schweizer et al., 2003). The Swiss Institute for Snow and Avalanche Research has reported an average of 22 fatalities caused by avalanches every year, with the number of non-fatal accidents being tenfold higher (SLF, 2025). These data show that avalanches are a significant threat, which is why it is crucial to understand how avalanches form and propagate in order to protect people and infrastructure better.

Avalanches are rapid masses of snow that move down mountain slopes due to gravity (Ancey, 2001). Snow avalanches can either be loose snow, which starts from a point in a cohesionless surface, wet snow, or snow slab avalanches (Schweizer et al., 2003). In this Master Thesis, the focus will be on dry snow slab avalanches, so mostly coherent masses of snow which contain no liquid water. Snow slab avalanches consist of a cohesive slab of snow (Schweizer et al., 2003) that is released because of loading at the surface, which triggers crack propagation in a weak layer (Gaume et al., 2017). Weak layers are characterized by the grain shape of snow, which leads to lower cohesion of those layers (Föhn et al., 1998).

As mentioned, avalanches are a major threat to human life in mountainous regions, which is why detecting the breakoff of the snow slab is important for hazard mitigation. Detecting the breakoff of the slab means that warnings can be issued in lower parts of the mountain that an avalanche is possible. Furthermore, detecting slab breakoff with DAS is also important to further understand crack dynamics in snow as well as the factors influencing crack propagation.

In this master's thesis, the aim is to detect avalanche slab breakoff and crack propagation using modeled data from distributed acoustic sensing (DAS) cables.

In this thesis, a fibre-optic cable installed on the slope is treated as a continuous seismic sensor, so that deformation can be observed along its full length rather than at only a few isolated stations (Parker et al., 2014). This is possible because DAS sends optical pulses through the fibre and records the backscattered light from every position along the cable (Shatalin et al., 1998). Ground motion near the cable, for example weak-layer failure or slab breakoff, changes the fibre strain, which appears as phase variations in the backscattered signal (Hartog, 2017). The phase variations are then converted into dynamic strain and from strain, into deformation along the cable (Trabattoni et al., 2023). mention also the cable location here, length etc

A pilot early-warning system using distributed acoustic sensing has been implemented in Norway, where avalanches are detected by their seismic signature (Turquet et al., 2024), meaning when the avalanche is already moving downhill. Such sensors are also located in Vallée de la Sionne in the Swiss Alps (Paitz et al., 2023). The detectability of avalanches by DAS has also been studied in the Swiss Alps by Paitz et al., 2023 and Edme et al., 2023, but the detectability of the slab breakoff itself has not been studied.

A multitude of models are used to study different processes in snow. The SNOWPACK and CROCUS models used in the Swiss and French alps model the structure and evolution of snow (Bartelt & Lehning, 2002; Brun et al., 1992). These models were developed to understand how snow evolves under different conditions, for example a changing temperature gradient and water content and what consequences this has for the formation of weak layers where failure likely occurs and avalanches are triggered (Bartelt & Lehning, 2002; Brun et al., 1992). SNOWPACK and CROCUS are comprehensive models that take into account multiple governing equations and processes and what consequences they have for the structure of snow. While it is important to understand the influences on for example temperature gradient on the snow structure and weak layer formation and stability as a consequence, this is not the focus of the thesis. This is because these models are too complex and comprehensive for the purpose of trying to detect slab breakoff in DAS data as they model the entire snow structure. Using SNOWPACK or CROCUS to investigate how different snowpack compositions and factors influence the DAS signal is beyond the scope of this thesis.

To model crack propagation in snow, multiple numerical modeling methods have been used. Cundall and Strack, 1979 developed the distinct element method (DEM) for granular media. This model discretizes

the domain as spheres and disks, which interact with each other based on Newton's second law (Cundall & Strack, 1979). The interactions of particles are modeled by the governing equations and the resulting forces and displacements (Cundall & Strack, 1979). This method has been used by Bobillier et al., 2020, 2024; Mulak and Gaume, 2019 and many others to model crack propagation and its influencing factors. The main limitation of DEM is that the particle contacts and the corresponding equations need to be recalculated every time a grain contact is made or broken (Cundall & Strack, 1979). This means that for large, slope-scale simulations, this method becomes increasingly computationally expensive. In addition to the high computation cost, DEM also cannot account for anticrack propagation, the closure of cracks in this propagation mode is accounted for by manually removing elements from the mesh, which is not very efficient (Gaume et al., 2019).

This is misleading. Salvo is not used for snow mechanics but for wave propagation. → "totally different" Physics

The same is true for finite element modelling (FE or FEM), which is done using SALVUS in this Thesis. FEM has been used to model snow mechanics since the 1970s, a review of the different approaches has been published by Podolskiy et al., 2013. The finite element modelling approach used in this thesis is described more in detail in ???. FE can also not account for anticrack propagation, similarly to DEM, mesh elements would need to be removed to account for anticrack propagation.

This problem is solved by using the material point method (MPM). In this method, the material is discretized in a set of Lagrangian points, which are used to track mass, deformation and momentum (Bardenhagen et al., 2000). Because the points are not fixed, an Eulerian background mesh is needed to calculate spatial derivatives (Bardenhagen et al., 2000). This method is not only more computationally efficient than DEM on the slope-scale, it can also account for anticrack behavior and strain softening of snow (Gaume et al., 2019). This is why the material point method will also be used in the thesis to investigate slab breakoff in DAS data on a large scale. The material point method has been used by Bergfeld et al., 2025, who modeled slope-scale experiments to understand the transition of crack propagation modes, Trotter et al., 2022 who modeled propagation saw tests to observe transitions in crack propagation speeds, as well as Gaume et al., 2019 and Gaume et al., 2018.

While MPM is very good at accounting for anticrack propagation, FE is more computationally efficient and accurate Lian et al., 2012. This is relevant especially for large-scale simulations, such as the ones done in this Thesis. This is because in MPM, the discretization through particles causes the displacement of each particle to be tracked and the contact at each particle to be calculated (Zhang et al., 2023). Furthermore, MPM does not include boundary conditions in the way FE does (Coombs et al., 2025). Finite element is more efficient on the slope scale because the mesh is fixed, as described by (Zienkiewicz et al., 2025). This means that the degrees of freedom and thus the number of calculations which need to be done are much smaller than in the MPM case. This means that MPM is more accurate at representing snow as a medium with its porous character and anticrack behavior, but FE is more efficient at performing large-scale simulations.

While it is important to understand the different modeling methods and their limitations, it is also important to understand the different types of crack propagation and what factors influence them. Understanding how fast a crack can propagate and which factors influence this is important for warning, as it relates the time of the initial crack to the time where the slab breaks off depending on snow and terrain parameters.

Two main types of crack propagation are believed to be important for weak layer failure: anticrack propagation and shear crack propagation (Bergfeld et al., 2025). There are three ways in which cracks can open up: mode I, where space is opened between the crack faces under tensile loading, and modes II and III, which represent parallel shearing and diagonal shearing (Heierli et al., 2003). In anticrack propagation, the weak layer consolidates under loading (Knight & Knight, 1972), while cracks propagate in a combination of modes I-III (Heierli et al., 2003). Gaume et al., 2018 modeled anticrack propagation induced by a propagation saw test (PST) using MPM, which showed that the scale of the simulations influences the propagation direction of the cracks. For smaller scale simulations, cracks propagate from top to bottom of a slope, whereas for slope-scale simulations, cracks propagate bottom to top, which means that remote triggering of avalanches is possible (Gaume et al., 2018). This is an important consideration for the thesis because it influences where the slab breakoff would be observed in terms of DAS cable location. Other models also show that the crack propagation speed of anticracks is influenced by slope angle and curvature: concave slope areas have higher propagation speeds than convex areas of

reference here

the slope (Gaume et al., 2019). Furthermore, there is a directionality in propagation speed, simulations by Gaume et al., 2019 show that cracks propagate twice as fast upslope than in the downslope direction, which is especially important when thinking about remote triggering. The slab breaks off when the slope angle exceeds the friction angle of the material (Gaume et al., 2019). Slope angle also influences which mode of crack propagation is more dominant (Rosendahl & Weißgraeber, 2020). In steeper terrains, shear modes (II and III) dominate, whereas in flatter terrain, mode I dominates the crack propagation (Rosendahl & Weißgraeber, 2020). The critical force, which is for example exerted by a skier, needed for weak layer failure is also dependent on slope angle; it decreases with increasing slope angle (Rosendahl & Weißgraeber, 2020). Snow parameters also have an influence on this critical force, as shown by simulations by Rosendahl and Weißgraeber, 2020: the thicker the slab or the higher the slab tensile strength, the better the force by the skier can be distributed on the weak layer and the more force is needed for failure.

Multiple experiments and numerical models have shown that cracks can propagate faster than the elastic wave speed in snow, so-called supershear crack propagation (Bergfeld et al., 2025; Bobillier et al., 2024; Trottet et al., 2022). Understanding what causes the transition from sub-Rayleigh to supershear crack propagation is of high importance for the thesis, because it can give an estimate of whether supershear crack propagation will occur under certain conditions, which leads to faster propagation of cracks and less time between loading and slab breakoff. In MPM models of PST, Trottet et al., 2022 observe that the occurrence of supershear cracks is dependent on slope angle: in steeper terrains, supershear propagation was observed. This can be explained by slab bending, because in steep terrains, bending of the slab is not limited to the collapse depth of the weak layer (Trottet et al., 2022). In in steeper terrain, the slab can bend more because of the terrain angle, the additional bending increases tensile stress in the slab, which leads to faster crack propagation (Trottet et al., 2022). These observations were also made by Bobillier et al., 2024 as well as Bergfeld et al., 2025. The release area of an avalanche is also tied to propagation speed: at supershear propagation speed, there is less crack arrest and therefore larger release areas (Bergfeld et al., 2025; Trottet et al., 2022). This is crucial because larger release areas also mean a larger snow volume, which has more potential for damage.

Independent of crack propagation mode, different factors influence the speed and distance at which cracks can propagate in the weak layer before failure occurs (Gaume et al., 2015, 2018; Gaume et al., 2017). As discussed above, slope angle and stiffness are important factors. Other factors include weak layer parameters: the weaker the weak layer, the slower crack propagation; the thicker the weak layer, the lower the crack propagation distance before failure (Gaume et al., 2015). The depth of the slab also has an influence on propagation distance and speed: the thicker the slab, the larger the propagation distance and velocity (Gaume et al., 2015). How fast a slab will break off is also dependent on the critical crack length needed for failure (Gaume et al., 2017). If the snowpack is more stable, which is the case when the slab has a high elastic modulus or the weak layer is thicker, the critical crack length is higher (Gaume et al., 2017) because the snow can sustain longer cracks before it fails. A very important factor in material strength is loading rate: if loading is slow, sintering can occur (Mulak & Gaume, 2019). This means that new snow contacts can form, which can strengthen the material (Mulak & Gaume, 2019). This is in contrast to rapid loading, which happens for example due to a skier (Mulak & Gaume, 2019).

In order to understand how crack propagation and slab breakoff would look in DAS data, numerical modeling of the wave propagation is done. This is done in SALVUS, which is a package that uses spectral elements to model seismic wavefields (Afanasyev et al., 2017).

The different snow processes described have different implications for the seismic wavefield, which will be modeled. Firstly, the loading which occurs before crack nucleation changes the stress state of the snowpack. The weak layer collapse and anticrack propagation produce volumetric compaction in the snowpack, this produces a seismic signal. The crack propagation, whether sub-Rayleigh or supershear, creates a moving rupture front, which can be likened to a moving source of seismic waves, the cracks itself also create a seismic signal. Lastly, the slab breakoff and sliding creates a seismic signal because a large mass of snow moves downhill, which likely creates the largest seismic signal of all of the processes mentioned. The signals of the different processes described will be modeled in the thesis, which will help to understand how the different processes look in DAS data and whether they can be distinguished from each other.

Based on the above information, there are multiple considerations necessary in order to model DAS data of crack propagation accurately. The scale of the models is an important factor, because as mentioned, it influences the direction of crack propagation. This influences at which location along or relative to the DAS cable slab breakoff will occur, which changes the measured data. Furthermore, multiple parameters influence the time passed between initial cracking and slab breakoff, these are important to consider when interpreting data. The different modes of crack propagation are also important to consider, it is possible that a transition between sub-Rayleigh and supershear crack propagation can be observed in the data. The loading rate can also be considered, because slower loading leads to stabilizing processes such as sintering.

This thesis will focus on the modelling of crack propagation, as recorded by DAS. Sub-Rayleigh as well as supershear crack propagation will be modelled using different methods and models. The aim is to determine what the different cases of crack propagation will look like in DAS data. This is then relevant for detecting actual avalanches in real-world data. The main questions are what crack propagation in different regimes looks like, how it differs when different modeling methods are used, and how the results can be explained in terms of wave physics.

3 Methods

3.1 Wave Propagation Modeling

The model described in ?? is implemented using the finite-element (FE) method in SALVUS. The following section describes the physics used by SALVUS. Understanding the physics behind the modeling has relevance for understanding and interpreting the results, but also what can be modeled. To discretize a domain, the domain is divided into a finite number of elements (Zienkiewicz et al., 2025). The number of elements in this case is defined by the maximum wave frequency that needs to be resolved accurately, the number of mesh elements per wavelength is user-input. The boundaries of these elements describe the mesh, the mesh has a finite number of grid-points (edge points of the elements) (Zienkiewicz et al., 2025). The governing equations are determined at every element node, so for rectangular elements

with nodes 1,2,3 and 4, there is the system $r^1 = \begin{pmatrix} r_1^1 \\ r_2^1 \\ r_3^1 \\ r_4^1 \end{pmatrix}$ for force and $u^1 = \begin{pmatrix} u_1^1 \\ u_2^1 \\ u_3^1 \\ u_4^1 \end{pmatrix}$ for displacement

(Zienkiewicz et al., 2025). The residual or characteristic relationship of linear elastic elements is of the form $r^1 = f^1 - K^1 u^1$, where f are the nodal forces distributing load on the nodes and K is the stiffness matrix of a specific element (Zienkiewicz et al., 2025). *Because in FE, the approximation of the wavefield by basis functions leads to a representation over the whole domain and not just at a set of points (Afanasiev et al., 2019). This means integrals can be calculated over surfaces or even volumes (Afanasiev et al., 2019).*

Snow is a porous medium (Mellor, 1977), which is approximated as a solid in this Thesis. This is because SALVUS meshes need an isotropic material and cannot account for pore spaces. Higher frequencies in the resulting data would be attenuated much quicker in an actual porous medium (**koushikDevelopmentSeismicSignalbase**). In addition, the presence of pore spaces in the medium would lead to complex scattering of the fields in the results, their absence is an important factor to consider when comparing simulated data to real-world data. The presence of air or water in snow is not neglected completely in the SALVUS simulations implemented however, because the material properties taken from Guillemot et al., 2021 are based on field measurements of snow. This means that the velocities measured by Guillemot et al., 2021 are not of pure snow, but snow that contains air or water.

The governing equation considered in this case is the elastic wave equation, because real-life crack propagation has P- and S-waves. The wave equation for a heterogenous elastic solid in the n-dimensional domain G is

$$\rho \partial_t^2 u = \nabla \cdot \sigma + f \quad (1)$$

(Afanasiev et al., 2019; Aki & Richards, 1981). Stress can be related to displacement using Hooke's law:

$\sigma = \frac{C}{\nabla u}$, where C is a 4th order tensor which characterizes the stiffness of the medium (Afanasiev et al., 2019). Combining ?? with Hoek's law, this leads to

$$\rho \partial_t^2 u = \nabla \cdot \frac{C}{\nabla u} + f, \quad (2)$$

which is the strong form of the linear elastic wave equation.

Boundary conditions, such as free surfaces, as well as initial conditions must be stated explicitly, they have the form $\sigma \cdot \hat{n}|_{\partial G} = \bar{\tau}(x), u|_{t=0} = \bar{u}(x), v|_{t=0} = \bar{v}(x)$ (Afanasiev et al., 2019). In this case, $\bar{\tau}(x), \bar{u}(x), \bar{v}(x)$ are the spatially variable distributions of traction, displacement and velocity (Afanasiev et al., 2019). The outward pointing normal vector on the boundary of domain G is described by $\hat{n}|_{\partial G}$.

FE methods use the weak form of the governing equations (Zienkiewicz et al., 2025). To obtain the weak form of the linear elastic wave equation in ??, the whole equation needs to be multiplied with a test function w, which is not time-dependent (Afanasiev et al., 2019). The weak form enforces the solution of the governing equation across the domain, instead of solving the equation everywhere, an integration over time leads to

$$\int_G \rho w \cdot \partial_t^2 u \, d^n x = \int_G w \cdot (\nabla \cdot \sigma) \, d^n x + \int_G w \cdot f \, d^n x \quad (3)$$

(Afanasiev et al., 2019). The divergence theorem relates the flux of a vector field through a closed surface to the divergence of the field in an enclosed volume: $\iiint_V (\nabla \cdot F) \, dV = \oint_{\partial V} (F \cdot \hat{n}) \, dS$, where F is the vector field. When this is applied to the first term on the right in ??, this leads to

$$\int_G \rho w \cdot \partial_t^2 u \, d^n x = \int_{\partial G} w \cdot (\sigma \cdot \hat{n}) \, d^{n-1} x - \int_G (\nabla w : \sigma) \, d^n x + \int_G w \cdot f \, d^n x \quad (4)$$

(Afanasiev et al., 2019). The first integral on the left side of ?? is the boundary integral, which changes depending on the boundary conditions implemented. The integral on the right hand side is the mass term, the second on the left hand side is the stiffness term, and the last on the left hand side is the external forcing term, all in ??. Equation ?? can be compacted to

$$(\rho \partial_t^2 u, w)_G - \langle \sigma \cdot \hat{n}, w \rangle_{\partial G} + (\sigma, \nabla w)_G = (f, w)_G, \quad (5)$$

using the notations $(a, b)_G = \int_G a \cdot b \, d^n x$, $\langle a, b \rangle_{\partial G} = \int_{\partial G} a \cdot b \, d^{n-1} x$ as well as Hoek's law (Afanasiev et al., 2019). This can be even more compacted by describing a mathematical model I with M for mass term, B for boundary term, S for stiffness and F for forcing term:

$$I_I(M) - I_I(B) + I_I(S) = I_I(F) = \sum_{P \in \{M, B, S, F\}} I_I(P) = 0 \quad (6)$$

(Afanasiev et al., 2019).

To solve ?? the Galerkin approach is used, the test function w and solution u are approximated by ϕ , which are a set of functions from the same space (Afanasiev et al., 2019). Any discrete function, for example displacement can be approximated this way:

$$g(x) \approx \sum_{j=1}^N F_j \phi_j(x), \quad (7)$$

where F_j is the expansion of the basis and g is the continuous function which is the approximation (Afanasiev et al., 2019). The derivatives of a function can then be approximated by the derivatives of ϕ .

To evaluate the integrals in ??, numerical quadrature is used Afanasiev et al., 2019. A certain number of integration weights W are combined, g(x) and the derivative of g(x) are evaluated at a set of integration points x_i (Afanasiev et al., 2019). This results in

$$\int_G g(x) \, d^n x \approx \sum_{i=1}^M W_i^G \sum_{j=1}^N F_j \phi_j(x_i) \quad (8)$$

well described, probably more detail than necessary

Hoek or Hoek?

(Afanasiev et al., 2019). In FE, an element-specific reference coordinate system θ is used. In this thesis, the domain consists of a snow layer and an air layer, which means there is a sharp discontinuity in model parameters. As Afanasiev et al., 2019 state, the model parameters and strain field are not differentiable. This is why a smooth polynomial basis of the governing equations over the whole domain is not appropriate, the basis B needs to consist of piecewise functions G defined on the individual elements (Afanasiev et al., 2019). The integration weights and basis function derivatives used, according to Afanasiev et al., 2019, are pre-tabulated in the element-specific coordinate system θ . To transform the integrals and derivatives to physical coordinates x , so the coordinates of the snow extent in this case, the element-specific Jacobian $\frac{\partial x}{\partial \theta}$ is used (Afanasiev et al., 2019). Equations ?? and ?? and their derivatives become:

$$g(x) \approx \sum_{j=1}^N F_j \phi_j(\theta(x)) \quad (9)$$

and

$$\nabla g(x) \approx \sum_{j=1}^N F_j \frac{\partial}{\partial \theta} \phi_j(\theta(x)) \frac{\partial \theta}{\partial x} \quad (10)$$

(Afanasiev et al., 2019). In equations ?? and ??, the function space V is approximated as $V_h \approx V$ in a previous step, this means the accuracy of the approximation is dependent on how well the original function space is approximated (Afanasiev et al., 2019). To make calculations over the whole domain G , the contributions from each element need to be summed:

$$\int_{G_e} g(x) d^n x \approx \sum_{i=1}^M W_i^{G_e} \sum_{j=1}^N F_j \phi_j(\theta(x_i)) \left| \frac{\partial x}{\partial \theta} \right|_{x=x_i}^n \quad (11)$$

for the volume and

$$\int_{\partial G_e} g(x) d^{n-1} x \approx \sum_{i=1}^M W_i^{\partial G_e} \sum_{j=1}^N F_j \phi_j(\theta(x_i)) \left| \frac{\partial x}{\partial \theta} \right|_{x=x_i}^{n-1} \quad (12)$$

for the surface of the domain (Afanasiev et al., 2019).

The dependence on the reference coordinate system is described by E , which is a model containing the physical coordinate to reference coordinate transformation, in other words, the determinant of the Jacobian matrix and its inverse (Afanasiev et al., 2019). Using basis B of the polynomial functions of the approximation V_h , which are piecewise per-element because of possible discontinuities, the abstract form in ?? becomes

$$\sum_{P \in \{M, B, S, F\}} I_I(P, B(E)) = 0 \quad (13)$$

(Afanasiev et al., 2019). This means that fields such as displacement and velocity are calculated in this way. From these fields, the data from the model in section ?? is obtained and plotted.

3.2 Model setups

The same two-layer medium as in section ?? was set up. In the simple case, a series of SALVUS moment tensor sources are set up with a certain distance. The time delay with which the sources fire is dependent on the sub-rayleigh or supershear case which is implemented. The bedrock or soil is not considered in both models. This is because preliminary tests of a 3-layered medium have shown that it acts as a transmitter of waves. For computational efficiency, the bedrock layer was thus neglected and the bottom of the snow was made a regular boundary, in other words, reflecting, to mimick the presence of a layer below the snow.

3.3 Cohesive Zone Model

The rupture model used in the simulations is implemented as a mixed-mode Cohesive Zone Model (CZM) following Mahajan and Joshi, 2008. This model was chosen because there is a narrow band ahead of the

fracture zone, the cohesive zone, where the material is already weakened. This means that before a point in the material cracks, there is already weakening of the material before. This seems to be reasonable for representing crack propagation in snow, since failure occurs as a consequence of loading and is not spontaneous. The computational domain extends 400 m in the horizontal (x) direction and 3 m in the vertical (y) direction and contains a 1.5 m snow layer over a 1.5 m air layer. Material parameters for the snow layer are taken from Guillemot et al., 2021. The mixed-mode behavior is implemented because in reality, crack propagation can involve both normal and shear displacement.

The mesh is generated to resolve frequencies up to the 95th percentile of the source wavelet's frequency content. Cohesive tractions are prescribed on zero-thickness interface elements and depend on the normalized opening $\xi = \frac{\delta}{\delta_c}$. The zero-thickness interface elements are implemented by not solving for volumetric strain but displacement between interfaces instead. For the linear softening law used here the traction in a given cohesive element follows

$$T(\xi) = T_{\text{peak}}(1 - \xi) \quad \text{for } 0 \leq \xi \leq 1, \quad T(\xi) = 0 \quad \text{for } \xi > 1,$$

where T_{peak} is the peak cohesive traction and δ_c the critical opening. The normalized opening can be related to the local crack coordinate as $\xi = \frac{(x-x_0)}{l_{\text{crack}}}$ for a crack front with initial position x_0 and local process length l_{crack} .

The mixed-mode potential used in the code follows the form from Mahajan and Joshi, 2008. With $\Delta = (\Delta_n, \Delta_t)$ denoting normal and tangential separations, the potential is written as

$$\phi(\Delta) = \phi_n \exp\left(-\frac{\Delta_n}{\delta_n}\right) \left\{ \left[1 + \frac{\Delta_n}{\delta_n}\right] + q \left[1 - \exp\left(-\frac{\Delta_t^2}{\delta_t^2}\right)\right] \right\},$$

from which the normal and tangential tractions are obtained, the implementation uses the same closed-form expressions. This is so that peak tractions occur at $\Delta_n = \delta_n$ and $|\Delta_t| = \delta_t/\sqrt{2}$. The normal and shear works of separation relate to peak strengths via

$$\phi_n = e \sigma_{\text{max}} \delta_n, \quad \phi_t = \sqrt{\frac{e}{2}} \tau_{\text{max}} \delta_t,$$

and the critical openings for pure modes are computed as

$$\delta_{n,c} = \frac{2G_I}{\sigma_n}, \quad \delta_{s,c} = \frac{2G_{II}}{\tau_s},$$

with the mixed-mode critical opening

$$\delta_c = \sqrt{(1 - r_{\text{mode}}) \delta_{n,c}^2 + r_{\text{mode}} \delta_{s,c}^2}.$$

Lamé parameters are derived from input seismic speeds and density: $v_s = \sqrt{\frac{\mu}{\rho}}$ so that $\mu = \rho v_s^2$, and $v_p = \sqrt{\frac{(\lambda+2\mu)}{\rho}}$ so that $\lambda = \rho(v_p^2 - 2v_s^2)$. Young's modulus is then computed from the Lamé parameters: $E = \frac{\mu(3\lambda + 2\mu)}{\lambda + \mu}$ (Aki & Richards, 1981). These conversions are applied for every material in the model.

The critical opening distances for pure normal (Mode I) and shear (Mode II) separations are computed from the corresponding fracture energies, strenghts, and peak tractions as

$$\delta_{n,c} = \frac{2G_I}{\sigma_n}, \quad \delta_{s,c} = \frac{2G_{II}}{\tau_s},$$

and combined for mixed-mode behaviour using

$$\delta_c = \sqrt{(1 - r_{\text{mode}}) \delta_{n,c}^2 + r_{\text{mode}} \delta_{s,c}^2},$$

where $r_{\text{mode}} \in [0, 1]$ controls the mode I/II mix, where 0 is pure mode I and 1 is pure mode II. In the current implementation the slip on each subfault follows the prescribed linear slip law up to the critical

slip distance. The mode mix is an user-input parameter, while all the other parameters are derived from laboratory experiments from Mahajan and Joshi, 2008.

The characteristic Irwin length for shear is calculated as $l_{ch} = \frac{E_{eff} G_{II}}{\tau_s^2}$, and the cohesive zone length is taken as $l_{cz} = \frac{\pi}{8} l_{ch}$, which sets the spatial extent over which cohesive weakening is active ahead of the crack tip.

Rupture propagation speed is chosen with respect to the local shear-wave speed and the bounds discussed in Trottet et al., 2022. Where appropriate, the rupture speed is corrected for finite slab thickness and bending stiffness to account for slab flexure effects. The rupture front is discretized into a sequence of subfaults: each subfault is activated at an onset time

$$t_o = \frac{x - x_{start}}{v_{rupture}},$$

using the subfault horizontal position x , the rupture starting position x_{start} , and the prescribed rupture speed $v_{rupture}$.

Each activated subfault is represented as a point/source in the SALVUS source array. The scalar seismic moment for a subfault is computed from the shear modulus, the prescribed slip, and the subfault width Δx :

$$M_0 = \mu x_{slip} \Delta x.$$

Moment-tensor components are set according to the user-specified mode-mix ratio, e.g. here the implementation uses $m_{xx} = 0$, $m_{xy} = M_0 r_{mode}$, $m_{yy} = -(1 - r_{mode})M_0$, which yields pure mode behaviour at the extremes of $r_{mode} = 0$ or 1, which means that an additional component of the moment tensor source becomes 0. All subfault sources use the same central frequency, wavelet shape, and amplitude scaling so the rupture is represented as a coherent moving source in the wave propagation simulations.

Key parameters and their provenance used in the simulations are summarised below:

Parameter	Typical value / source
σ_n	0.00184 MPa from Mahajan and Joshi, 2008
τ_s	0.004 MPa from Mahajan and Joshi, 2008
G_I, G_{II}	0.35, 0.50 J/m ² from Mahajan and Joshi, 2008
v_s, v_p, ρ	150, 332 m/s from Guillemot et al., 2021 (converted to μ, λ, E)
r_{mode}	User input (0 = Mode I, 1 = Mode II)

3.4 MPM PSD test in FE

- How exactly is the mpm data processed for fe input????
- The different components resulting from the MPM data are weighted in arrays and then translated into SALVUS sources. This is done by defining weights for every component in every source location. These are then passed to custom source time functions in SALVUS, where the mxx, myy and mxy components of every wavelet at the source location is weighted.

4 Results

- For all cases show: **Space time plots:** Strain (x) rate plots, Strain (x), velocity, in all cases at crack depth and snow bottom. **FK plots with velocites drawn, wiggle plots or single traces,** possibly PSD and STF plots to support certain cases, but i'm not sure these are needed because they look similar in all cases?
- Put animations via link, stills of all animation cases

- ② • Results of Complex Model (Cohesive zone) in FE, supersher vs sub-rayleigh, possibly weak layer vs no weak layer
- Results for simple FE model, all the same comparisons ①
- ③ • Results of MPM-based model, all the same comparisons
- MAYBE: ~~show some tests of weighted / custom stf so process can be followed better~~
technical detail → only describe the one we show!

5 Discussion

- Compare results of all models with wave physics (e.g what equations can explain which phenomena and interactions of the waves)
- Lamb waves physics
- Timoshenko beam effects MAYBE???? *→ rather not, only if you are convinced*
- FE vs MPM model discussion explained by different methodology, or find another way to explain differences
- I would like to lead from my input wave equation to the resulting waves observed in the results in a mathematical way, eg show the input equation, all the interactions and then the resulting equations of for example lamb waves:
- From the equation of motion in an elastic solid, which is

$$\mu \nabla^2 u + (\lambda + \mu) \nabla (\nabla \cdot u) = \rho \frac{\partial^2 u}{\partial t^2}, \quad (14)$$

the scalar and vector potential for P- and S-waves (Potentials ϕ, ψ) can be derived using the Helmholtz-Decomposition and the displacement becomes

$$u = \nabla \phi + \nabla \times \psi. \quad (15)$$

Splitting this into the horizontal and vertical component of displacement:

$$u_x = \frac{\partial \phi}{\partial x} - \frac{\partial \psi}{\partial z} \quad (16)$$

and

$$u_z = \frac{\partial \phi}{\partial z} + \frac{\partial \psi}{\partial x}. \quad (17)$$

When these are substituted into ??, it becomes

$$\nabla^2 \phi = \frac{1}{c_p^2} \frac{\partial^2 \phi}{\partial t^2}, \quad (18)$$

for P-waves and

$$\nabla^2 \psi = \frac{1}{c_s^2} \frac{\partial^2 \psi}{\partial t^2} \quad (19)$$

for S-waves, the wave speeds are derived from the Lamé-Parameters, as described in ??.

When assuming harmonic plane wave solutions for waves with wavenumber k and frequency ω , the harmonic solutions have the form

$$\phi(x, z, t) = f(z) e^{i(kx - \omega t)} \quad (20)$$

and

$$\psi(x, z, t) = g(z) e^{i(kx - \omega t)} \quad (21)$$

. When these are substituted into the wave equations, $f(z)$ and $g(z)$ yield ordinary differential equations, which are

$$\frac{d^2 f}{dz^2} + p^2 f = 0 \quad (22)$$

and

$$\frac{d^2 g}{dz^2} + q^2 g = 0, \quad (23)$$

with $p^2 = \frac{\omega^2}{c_p^2} - k^2$ and $q^2 = \frac{\omega^2}{c_s^2} - k^2$. These equations lead to the general solutions

$$\phi = [A \cos(pz) + B \sin(pz)] e^{i(kx - \omega t)} \quad (24)$$

and

$$\psi = [C \cos(qz) + D \sin(qz)] e^{i(kx - \omega t)}. \quad (25)$$

At the snow-air interface, there is a stress-free boundary because of the impedance contrast, according to **igelComputationalSeismologyPractical2016**. This means that both the normal and the shear stress need to be zero:

$$\sigma_{zz} = \lambda \nabla^2 \phi + 2\mu \frac{\partial u_z}{\partial z} = 0 \quad (26)$$

and

$$\sigma_{xz} = \mu \left(\frac{\partial u_x}{\partial z} + \frac{\partial u_z}{\partial x} \right) = 0. \quad (27)$$

These can be expressed in terms of the scalar and vector potential using the displacement potentials.

$$\sigma_{zz} = \lambda \left(\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial z^2} \right) + 2\mu \left(\frac{\partial^2 \phi}{\partial z^2} + \frac{\partial^2 \psi}{\partial x \partial z} \right), \quad (28)$$

$$\sigma_{xz} = \mu \left(\frac{\partial^2 \phi}{\partial x \partial z} - \frac{\partial^2 \psi}{\partial z^2} + \frac{\partial^2 \phi}{\partial x \partial z} + \frac{\partial^2 \psi}{\partial x^2} \right), \quad (29)$$

of which the first terms can be substituted with the wavespeed using the wavenumber-frequency and velocity relationship from the Lamé parameters. This results in

$$\sigma_{zz} = \mu [(q^2 - k^2)\phi + \frac{\partial^2 \psi}{\partial x \partial z}] \quad (30)$$

and

$$\sigma_{xz} = \mu [2 \frac{\partial^2 \phi}{\partial z} + \frac{\partial^2 \psi}{\partial x^2} - \frac{\partial^2 \psi}{\partial z^2}], \quad (31)$$

from which harmonic solutions containing sin and cos can be obtained. Here, a derivative is a multiplication with ik , which leads to

$$\sigma_{zz} = \mu [(q^2 - k^2)\phi + 2ik \frac{\partial \psi}{\partial z}] \quad (32)$$

and

$$\sigma_{xz} = \mu [2ik \frac{\partial \phi}{\partial z} + (q^2 - k^2)\psi]. \quad (33)$$

When taking into account the stress free boundary and substituting $K = (q^2 - k^2)$, then the normal stress at $\pm z$ becomes

$$\sigma_n = K[A \cos(pz) + B \sin(pz) + 2ik(-q(\sin(qz) + qD \cos(pz)))] = 0 \quad (34)$$

and for the shear stress

$$\sigma_s = 2ik[-pA \sin(pz) + pB \cos(pz)] + K[C \cos(qz) + D \sin(qz)] = 0 \quad (35)$$

at the top and bottom boundary of the snow layer. The matrix for symmetric Lamb waves is obtained by adding the top and bottom term of the normal stress and subtracting the shear stresses. This leads to

$$2K A \cos(pz) + 4ikq D \cos(qz) = \quad (36)$$

and

$$-4ikpA\sin(pz) + 2KD\sin(qz) = \quad (37)$$

, which can be written as a matrix:

$$\begin{pmatrix} K\cos(pz) & 2ikq\cos(qz) \\ -2ikp\sin(pz) & K\sin(qz) \end{pmatrix} \begin{pmatrix} A \\ D \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (38)$$

If the determinant of this matrix is set to zero, this results in

$$\frac{\tan(qz)}{\tan(pz)} = -\frac{2k^2pq}{(q^2 - k^2)^2} \quad (39)$$

, which is the dispersion of the symmetric mode of the Lamb waves, as described by Su and Ye, 2009. The same can be done for the asymmetric Lamb wave mode if the operations of the stresses are opposite.

Lamb waves are elastic guided waves that occur in plates (Su & Ye, 2009). They consist of particle motion in the direction of the wave polarization as well as the direction perpendicular to it (Su & Ye, 2009). They are guided by the boundaries of the layer or media in which they propagate and have a symmetric and an asymmetric mode (Su & Ye, 2009). The symmetric mode dominates the in-plate motion, while the antisymmetric mode dominates the vertical (out of plane motion) (Su & Ye, 2009).

- Compare a figure of lamb wave propagation of the different modes to the results obtained in GIFS to clearly show that they are present:
- In space time plots: describe the different wavefronts which can be seen. In sub-rayleigh, slope of stondest main front is sources firing (same as propagation speed), wavefront which travels ahead of this is smaller. In supershear, only one distinct front and backwards propagation: WHICH TYPES OF WAVES ARE THEY AND WHY
- Possibly link different observations to physical processes (eg leading wavefronts) that could be useful in detecting slab breakoff in real-world data *if you have any cases, better circulate on a few cases.*
- Also show fk plots of all the case and discuss which speed dominates when (eg in sub-rayleigh it's the slab speed) → possibly timoshenko effects here because of the 4th order term when frequency goes higher approximates medium speed? I think i'll explain this primarily with the highest speed present: in the sub-rayleigh case, the p-wave speed of the slab is the highest speed in the system, in the supershear case, the highest speed present is the crack propagation speed. *→ no, crack is still slower than the p-wave!*
- I would like to discuss the implications of the different wave types: eg have a clear, physical explanation for what different types of waves mean and how this is shown in terms of the crack or the snowpack → the lamb waves are caused by the impedance contrast at snow-air causing a stress free-boundary, which then guides the waves (link to derivation) *→ yes, that is good. Not sure deriving the full math is necessary in the main body (maybe appendix)*
- Tie all of this to literature of fracture mechanics and avalanche modelling, to also show where the phenomena come from and they could for example show up in not seismic data
- Important caveat: different stages of crack propagation as described by Bobillier et al., 2024 are irrelevant here, because crack propagates at constant speed → this is also not very physical and an important consideration for comparison to data as well as interpretation *→ yes, this is of course different once we lie*

6 Conclusion

- Link between all different models and scenarios and big takeaways for DAS data
- Summary of observations, compare and contrast models
- What can be built on this? eg more accurate Salvus model with multiple layers and topography etc

(3D)

- also add cracks not at steady state speed: crack propagation in transition mechanisms described in introduction
- Slope effects will probably change results a lot
- It would be nice to have a "how to detect slab breakoff in DAS data" here, for example what types of waves to look for, what order they have etc

7 Notes for Results & Discussion

7.1 Slab Mechanics

Siron et al., 2023 model their snow slab as a Timoshenko beam. Such beams are thick beams, sandwich composite beams, or beams where the wavelength is similar to the beam thickness (Siron et al., 2023). With a central frequency of $f_0 = 20\text{Hz}$, the wavelength of the source used in this thesis is $\nu = \frac{\lambda}{v}$, where ν is the frequency and v is the speed of the medium. This leads to a central frequency wavelength of 15m for the shear wave and 30m for the compressional wave, this means the snow slab layer can be seen as a Timoshenko beam. According to Mei and Mace, 2004, Timoshenko beams act as waveguides. In contrast to Euler-Bernoulli models, which only consider lateral inertia and elastic forces caused by bending deflections, Timoshenko beams also consider rotary inertia and shear distortion (Mei & Mace, 2004). This is relevant for snow slabs because they can be distorted under loading which can cause avalanches. The equation of motions for a beam element along the x-axis in a Timoshenko beam are:

$$GA\kappa\left[\frac{\partial\psi(x,t)}{\partial x} - \frac{\partial^2 y(x,t)}{\partial x^2}\right] + \rho A \frac{\partial^2 y(x,t)}{\partial t^2} = q(x,t) \quad (40)$$

and

$$EI \frac{\partial^2 \psi(x,t)}{\partial x^2} + GA\kappa\left[\frac{\partial y(x,t)}{\partial x} - \psi(x,t)\right] - \rho I \frac{\partial^2 y(x,t)}{\partial t^2} = 0 \quad (41)$$

(Mei & Mace, 2004). In equations ?? $\frac{\partial y(x,t)}{\partial x}$, $\frac{\partial y(x,t)}{\partial x} - \psi(x,t)$ are the slope of the center axis of the beam and the shear angle (Mei & Mace, 2004). Both equations of motion are coupled through the slope and the transversal (across) deflexion of the beam, the shear force and bending moment are also related to these two (Mei & Mace, 2004). The shear force $V(x,t)$ is related according to

$$V(x,t) = GA\kappa\left[\frac{\partial y(x,t)}{\partial x} - \psi(x,t)\right] \quad (42)$$

and the bending moment $M(x,t)$ according to

$$M(x,t) = -EI \frac{\partial \psi(x,t)}{\partial x} \quad (43)$$

(Mei & Mace, 2004).

The propagation matrices of sources at points A and B can be described by the vectors of wave propagation in positive and negative direction of propagation a^+, a^-, b^+, b^- (Mei & Mace, 2004). Using the propagation matrix for distance x which is $f(x) = \begin{bmatrix} e^{-ik_1 x} & 0 \\ 0 & e^{-ik_2 x} \end{bmatrix}$, the vectors can be related according to $a^- = f(x)b^-, b^+ = f(x)a^+$ (Mei & Mace, 2004). A similar matrix can be obtained for the reflections at boundaries, assuming equilibrium of shear and bending at the boundary (Mei & Mace, 2004). This can be considered as

$$-V = HA\kappa\left(\frac{\partial y}{\partial x}\psi\right) = \bar{K}_{TY} + \bar{K}_{TR}\psi \quad (44)$$

and

$$M = -EI \frac{\partial y}{\partial x} = \bar{K}_{R\psi} + \bar{K}_{RTY} \quad (45)$$

(Mei & Mace, 2004). This means that both incident propagating and transitional wave modes are fully reflected, keeping their original wave types but with a 180 degree phase change (Mei & Mace, 2004). This is important to consider when looking at results of reflections in the snow slab. At a free surface, which the snow-air interface is, the phase angle of the reflected wave is frequency dependent, and a propagation mode arises from the incident transition mode (Mei & Mace, 2004).

→ this is interesting, but what do we do with it? Does it help modeling or interpretation? Does it fit in your story? If not it might be an outlook or appendix.

7.2 Lamb Wave propagation Physics

Lamb waves are elastic guided waves that occur in plates (Su & Ye, 2009). They consist of particle motion in the direction of the wave polarization as well as the direction perpendicular to it (Su & Ye, 2009). They are guided by the boundaries of the layer or media in which they propagate and have a symmetric and an asymmetric mode (Su & Ye, 2009). The symmetric mode dominates the in-plate motion, while the antisymmetric mode dominates the vertical (out of plane motion) (Su & Ye, 2009). Lamb waves are dispersive. Lamb waves in a plate with thickness d and half thickness h can be described as

$$\frac{\tan(qh)}{\tan(ph)} = \frac{4k^2qp\mu}{(\lambda k^2 + \lambda p^2 + 2\mu p^2)(k^2 - q^2)} \quad (46)$$

(Su & Ye, 2009). This can be split up into

$$\frac{\tan(qh)}{\tan(ph)} = -\frac{4k^2qp}{(k^2 - q^2)^2} \quad (47)$$

for symmetric and

$$\frac{\tan(qh)}{\tan(ph)} = -\frac{(k^2 - q^2)^2}{4k^2qp} \quad (48)$$

for antisymmetric modes (Su & Ye, 2009). In isotropic materials these equations are

$$\frac{\tan(qh)}{q} = \frac{4k^2p \tan(ph)}{(k^2 - q^2)^2} = 0 \quad (49)$$

in the symmetric case and

$$q \tan(qh) + \frac{(k^2 - q^2)^2 \tan(ph)}{4k^2p} = 0 \quad (50)$$

for the antisymmetric waves (Su & Ye, 2009). These are also the dispersion equations of the two Lamb wave modes, the symmetrical mode is more dispersive than the antisymmetric mode because of the division by q in equation ?? (Su & Ye, 2009).

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